

Neurosurgical Anesthesia MCQ – Chapter 27

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 27)

60 Questions • 35 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is cerebral perfusion pressure (CPP)?

- A. Sum of MAP and ICP
- B. Difference between CVP and ICP
- C. Difference between mean arterial pressure and intracranial pressure
- D. Difference between systolic and diastolic pressure

Ans: C

Q2. What is the normal cerebral blood flow (CBF) per 100 g of brain tissue?

- A. 75 mL
- B. 25 mL
- C. 50 mL
- D. 100 mL

Ans: C

Q3. Between what range of mean arterial pressure is CBF maintained at constant level by autoregulation?

- A. 50 and 100 mm Hg
- B. 60 and 150 mm Hg
- C. 80 and 180 mm Hg
- D. 40 and 120 mm Hg

Ans: B

Q4. Which of the following can abolish cerebral autoregulatory mechanism?

- A. Moderate hypocapnia
- B. Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents
- C. Propofol
- D. Mild hypothermia

Ans: B

Q5. By how much does each mm increase in pCO₂ above 40 mm Hg increase CBF?

- A. 5-10 mL/min
- B. 15-30 mL/min
- C. 50-70 mL/min
- D. 35-50 mL/min

Ans: B

Q6. When does pO₂ increase cerebral blood flow?

- A. Only if pO₂ is below 60 mm Hg
- B. When pO₂ is between 60-80 mm Hg
- C. When pO₂ is above 100 mm Hg
- D. Always

Ans: A

Q7. Which IV anesthetics decrease cerebral blood flow?

- A. All agents decrease CBF
- B. All inhalational agents, ketamine and opioids increase CBF while barbiturates, propofol, etomidate and benzodiazepines decrease CBF
- C. Only barbiturates
- D. Ketamine and opioids

Ans: B

Q8. What increases the cerebral metabolic rate (CMR)? A. Barbiturates B. Propofol C. Hypothermia and anesthetics D. Hyperthermia and seizures	Ans: D
Q9. What is the normal intracranial tension (ICT)? A. 20-25 mm Hg B. 10-15 mm Hg C. 5-8 mm Hg D. 1-5 mm Hg	Ans: B
Q10. At what level does ICT become significant and need aggressive treatment? A. > 20 mm Hg B. > 15 mm Hg C. > 30 mm Hg D. > 25 mm Hg	Ans: A
Q11. What are the signs of raised ICT? A. Headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death B. Only papilledema C. Only headache D. Tachycardia and hypotension	Ans: A
Q12. What midline shift on CT scan indicates very high ICT? A. > 0.5 cm B. > 1.0 cm C. > 1.5 cm D. > 0.2 cm	Ans: A
Q13. What is the most rapid method of reducing ICT? A. Mannitol B. Steroids C. Diuretics D. Hyperventilation	Ans: D
Q14. What is the recommended pCO₂ range to maintain during hyperventilation for reducing ICT? A. 25-30 mm Hg B. 35-40 mm Hg C. 30-35 mm Hg D. 20-25 mm Hg	Ans: A

Q15. What is the most commonly used hyperosmotic drug for reducing ICT? A. Urea B. Mannitol C. Glycerol D. Hypertonic saline	Ans: B
Q16. Why should the use of mannitol in head injury and stroke be judicious? A. It causes hyperglycemia B. It has no effect C. If blood brain barrier is disrupted, mannitol can enter brain tissue increasing cerebral edema D. It causes hyponatremia	Ans: C
Q17. Which diuretic is very effective for reducing ICT particularly when blood brain barrier is disrupted? A. Hydrochlorothiazide B. Spironolactone C. Mannitol D. Furosemide	Ans: D
Q18. How do barbiturates (Thiopentone) reduce ICT? A. By increasing CMR B. By causing cerebral vasoconstriction which decreases CSF production C. By increasing CBF D. By causing vasodilation	Ans: B
Q19. Which posture helps decrease ICT by increasing venous drainage? A. Head up position B. Supine position C. Prone position D. Trendelenburg position	Ans: A
Q20. What can increase ICT by compressing jugular veins? A. Neck flexion B. Lateral rotation C. Extension D. Head up position	Ans: A
Q21. If ICT is not controlled by conservative measures, how should CSF be drained? A. Through lumbar puncture B. Through subdural space C. Through ventricles (never through lumbar puncture as brain herniation can cause death) D. Through epidural space	Ans: C

Q22. What is the effect of all inhalational agents on cerebral blood flow and ICT? A. No effect B. All inhalational agents (including nitrous oxide) increase CBF and hence ICT C. All decrease CBF and ICT D. Only halothane increases CBF	Ans: B
Q23. What is the order of vasodilating potency of inhalational agents? A. Halothane >> Desflurane ≈ Isoflurane > Sevoflurane B. Sevoflurane > Isoflurane > Desflurane > Halothane C. Desflurane > Halothane > Isoflurane > Sevoflurane D. All are equal	Ans: A
Q24. By how much can Isoflurane and Sevoflurane decrease cerebral metabolic rate (CMR)? A. Up to 50% B. Up to 100% C. Up to 25% D. Up to 75%	Ans: A
Q25. Which inhalational agent is preferred for neurosurgery in current day practice? A. Halothane B. Isoflurane C. Desflurane D. Sevoflurane	Ans: D
Q26. Which IV anesthetics are most potent in decreasing CBF, CMR, CMO₂ and ICT? A. Opioids B. Barbiturates C. Ketamine D. Propofol	Ans: B
Q27. What is the effect of opioids on CBF in neurosurgical patients? A. Decrease CMR and CMO₂ but increase CBF (directly acting cerebral vasodilator) and hence ICT B. Decrease CBF and ICT C. No effect D. Increase CMR	Ans: A
Q28. By how much can Ketamine increase CBF and ICT? A. Up to 20% B. Up to 90% C. Up to 30% D. Up to 60%	Ans: D

Q29. What is the effect of succinylcholine on ICT? A. Increases ICT B. No effect C. Variable effect D. Decreases ICT	Ans: A
Q30. Do non-depolarizing muscle relaxants cross the blood brain barrier? A. Only vecuronium B. No, they do not cross the blood brain barrier C. Yes, all of them D. Only pancuronium	Ans: B
Q31. What is the effect of lignocaine on CBF, CMR and CMO2? A. Increases all B. Decreases CBF, CMR, CMO2 (but at toxic doses can induce seizure and increase CMR) C. Only decreases CMR D. No effect	Ans: B
Q32. How do vasopressors (Mephentermine, Ephedrine, Epinephrine) affect CBF? A. No effect on CBF B. Directly increase CBF C. Increase CBF indirectly by increasing cerebral perfusion pressure D. Decrease CBF	Ans: C
Q33. Why should glucose containing solutions be avoided in neurosurgery? A. Hyperglycemia can precipitate cerebral edema B. They cause hypernatremia C. They are too viscous D. They cause hypoglycemia	Ans: A
Q34. What is the best policy for fluid management in neurosurgical patients? A. Only Ringer lactate B. Alternate ringer and normal saline liter by liter C. Only normal saline D. Only colloids	Ans: B
Q35. Is routine use of hypothermia recommended in neurosurgery? A. Yes, always B. No, routine use of hypothermia is not recommended C. Only in spine surgery D. Only in posterior fossa surgery	Ans: B
Q36. What is the optimal blood sugar range to maintain in neurosurgical patients? A. 80-120 mg% B. 100-140 mg% C. 140-180 mg% D. 180-220 mg%	Ans: C

Q37. What is the most important monitor for neurosurgical patients? A. Capnography B. ECG C. Temperature D. Pulse oximetry	Ans: A
Q38. What is the beneficial range of pCO₂ during monitoring in neurosurgery? A. 40-45 mm Hg B. 20-25 mm Hg C. 35-40 mm Hg D. 25-30 mm Hg (moderate hypocapnia)	Ans: D
Q39. Why should hypoxia be avoided during pulse oximetry monitoring in neurosurgery? A. It causes hypothermia B. It causes bradycardia C. It decreases CMR D. It increases ICT by causing cerebral vasodilation	Ans: D
Q40. What is essential for smooth emergence in neurosurgical patients? A. Emergence free of coughing, straining and hypertension B. Use of succinylcholine C. Rapid extubation D. Deep extubation always	Ans: A
Q41. What is the best method to achieve smooth emergence in neurosurgery? A. Give succinylcholine B. Deep extubation C. Inject lignocaine before extubation, continue inhalational (at least N₂O) till dressing and use beta blockers D. Only give opioids	Ans: C
Q42. What is the target CPP (MAP-ICP) to maintain during neurosurgery? A. 100-120 mm Hg B. 40-60 mm Hg C. 70-100 mm Hg D. 50-70 mm Hg	Ans: C
Q43. Are central neuraxial blocks contraindicated in raised ICT? A. They are absolutely contraindicated B. They are safe C. Only epidural is contraindicated D. They are relatively contraindicated	Ans: A

Q44. Which induction agent is preferred over thiopentone for neurosurgery? A. Midazolam B. Propofol C. Etomidate D. Ketamine	Ans: B
Q45. Why can intubation be problematic in neurosurgical patients? A. Intubation can significantly increase ICT due to response to laryngoscopy B. Vocal cord paralysis C. Difficult anatomy D. Risk of aspiration	Ans: A
Q46. Is Suxamethonium absolutely contraindicated for neurosurgery? A. Only in posterior fossa surgery B. Yes, absolutely contraindicated C. Not absolutely contraindicated - can be safely used for rapid sequence induction or difficult airway D. Only in spine surgery	Ans: C
Q47. What agents should be used for maintenance if ICP is too high and surgical field too tight? A. Halothane B. Nitrous oxide and sevoflurane C. High concentration inhalational agents D. IV agents only - inhalational agents should be avoided	Ans: D
Q48. What should be assumed about all head injury patients regarding cervical injury? A. Only in high-velocity trauma B. Only in elderly patients C. Cervical injury is unlikely D. All patients should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise	Ans: D
Q49. When should nasal intubation NOT be performed in head injury? A. In all head injury patients B. Only in pediatric patients C. When GCS is above 10 D. Unless basal skull fracture has been ruled out	Ans: D
Q50. What is the best method for assessing neurologic status in head injury? A. Modified Glasgow Coma Scale B. MRI C. EEG D. CT scan	Ans: A

Q51. What is the normal Glasgow Coma Scale score? A. 10/10 B. 15/15 C. 12/12 D. 20/20	Ans: B
Q52. A GCS score of 7 or less persisting for more than 6 hours is associated with what mortality? A. More than 10% B. More than 20% C. More than 35% D. More than 50%	Ans: C
Q53. Why is DIC commonly associated with head injury? A. Due to hypoxia B. Due to blood loss C. Due to hypothermia D. Due to release of brain thromboplastin	Ans: D
Q54. What is the most sensitive tool to detect venous air embolism (VAE)? A. CVP monitoring B. End tidal CO₂ C. Precordial Doppler D. Transesophageal echocardiography (TEE)	Ans: D
Q55. At what volume of entrained air does VAE become clinically significant? A. > 25 mL B. > 200 mL C. > 100 mL D. > 50 mL	Ans: C
Q56. What is the incidence of VAE in posterior fossa craniotomies? A. 20-40% B. 5-10% C. 1-5% D. 50-60%	Ans: A
Q57. What is the first step in treatment of venous air embolism? A. Start CPR B. Stop nitrous oxide and start 100% oxygen (N₂O can expand the size of air embolus) C. Give vasopressors D. Trendelenburg position	Ans: B

Q58. What is the sensitivity order for detection of air embolism in decreasing order?

- A. Clinical signs > ECG > CVP > TEE
- B. ECG > CVP > TEE > Doppler
- C. Doppler > ECG > CVP > TEE
- D. TEE > Precordial Doppler > PAP, ETCO₂, ET nitrogen > CVP > ECG > Clinical signs

Ans: D

Q59. What are the clinical signs of venous air embolism?

- A. Hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse
- B. Fever and tachypnea
- C. Only hypotension
- D. Hypertension and bradycardia

Ans: A

Q60. After stroke surgery, when is the period of maximal vasospasm risk that should be avoided?

- A. 4-14 days
- B. 1-3 days
- C. 15-21 days
- D. 21-30 days

Ans: A

Answer Key & Explanations

Q1. Answer: C — Difference between mean arterial pressure and intracranial pressure

Cerebral perfusion pressure (CPP) is the difference between mean arterial pressure and intracranial pressure.

Topic: Cerebral Physiology

Q2. Answer: C — 50 mL

CBF is 50 mL/100 g of brain tissue.

Topic: Cerebral Physiology

Q3. Answer: B — 60 and 150 mm Hg

CBF is maintained at constant level by autoregulation when mean arterial pressure is between 60 mm Hg and 150 mm Hg.

Topic: Cerebral Physiology

Q4. Answer: B — Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents

Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents can abolish this autoregulatory mechanism.

Topic: Cerebral Physiology

Q5. Answer: B — 15-30 mL/min

Each mm increase in pCO₂ above 40 mm Hg increases CBF by 15-30 mL/min.

Topic: Cerebral Physiology

Q6. Answer: A — Only if pO₂ is below 60 mm Hg

pO₂ increases CBF only if pO₂ is below 60 mm Hg.

Topic: Cerebral Physiology

Q7. Answer: B — All inhalational agents, ketamine and opioids increase CBF while barbiturates, propofol, etomidate and benzodiazepines decrease CBF

All inhalational agents, ketamine and opioids increase CBF while propofol, etomidate and benzodiazepines decreases the CBF.

Topic: Cerebral Physiology

Q8. Answer: D — Hyperthermia and seizures

Hyperthermia, seizures increase the cerebral metabolic rate (CMR) while hypothermia and anesthetics decreases the cerebral metabolic rate.

Topic: Cerebral Physiology

Q9. Answer: B — 10-15 mm Hg

Normal ICT is 10-15 mm Hg.

Topic: Cerebral Physiology

Q10. Answer: A — > 20 mm Hg

ICT > 20 mm Hg is significant and needs aggressive treatment.

Topic: Cerebral Physiology

Q11. Answer: A — Headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death

Signs include headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death.

Topic: Cerebral Physiology

Q12. Answer: A — > 0.5 cm

At very high ICT, midline shift > 0.5 cm on computed tomography (CT) scan is seen.

Topic: Cerebral Physiology

Q13. Answer: D — Hyperventilation

Hyperventilation is the most rapid method of reducing ICT. It reduces ICT by causing hypocapnia which causes cerebral vasoconstriction to decrease CSF production.

Topic: Methods to Decrease ICT

Q14. Answer: A — 25-30 mm Hg

It is recommended to maintain pCO₂ between 25-30 mm Hg. Excessive hypocapnia (< 25 mm Hg) can cause cerebral ischemia.

Topic: Methods to Decrease ICT

Q15. Answer: B — Mannitol

Among hyperosmotic agents, mannitol is most commonly used. These drugs are retained in intravascular compartment increasing plasma osmolarity, drawing water from brain tissue.

Topic: Methods to Decrease ICT

Q16. Answer: C — If blood brain barrier is disrupted, mannitol can enter brain tissue increasing cerebral edema

If blood brain barrier is disrupted (head injuries, stroke) these agents can enter the brain tissue increasing the cerebral edema. Therefore, the use of mannitol in head injury and stroke should be judicious.

Topic: Methods to Decrease ICT

Q17. Answer: D — Furosemide

Furosemide is a very effective method of reducing ICT particularly when blood brain barrier is disrupted.

Topic: Methods to Decrease ICT

Q18. Answer: B — By causing cerebral vasoconstriction which decreases CSF production

Barbiturates: Thiopentone by causing cerebral vasoconstriction is effective in decreasing the production of CSF and hence ICT.

Topic: Methods to Decrease ICT

Q19. Answer: A — Head up position

Head up position by increasing venous drainage decreases ICT.

Topic: Methods to Decrease ICT

Q20. Answer: A — Neck flexion

Neck flexion by compressing jugular veins can increase ICT.

Topic: Methods to Decrease ICT

Q21. Answer: C — Through ventricles (never through lumbar puncture as brain herniation can cause death)

CSF drainage: If ICT is not controlled by conservative measure it should be drained through ventricles (never through lumbar puncture, brain herniation can even cause death).

Topic: Methods to Decrease ICT

Q22. Answer: B — All inhalational agents (including nitrous oxide) increase CBF and hence ICT

All inhalational agents (including nitrous oxide) increases the cerebral blood flow and hence ICT.

Topic: Pharmacology

Q23. Answer: A — Halothane >> Desflurane ≈ Isoflurane > Sevoflurane

The order of vasodilating potency is halothane >> Desflurane ≈ Isoflurane > Sevoflurane.

Topic: Pharmacology

Q24. Answer: A — Up to 50%

Isoflurane and Sevoflurane can decrease CMR up to 50%.

Topic: Pharmacology

Q25. Answer: D — Sevoflurane

Due to better preservation of autoregulation, equivalent reduction in CMR and smoother recovery, sevoflurane is preferred over isoflurane as an inhalational agent of choice for neurosurgery.

Topic: Pharmacology

Q26. Answer: B — Barbiturates

Barbiturates, propofol, etomidate and benzodiazepines decrease CBF, CMR, CMO₂ and ICT. Barbiturates are most potent in this aspect.

Topic: Pharmacology

Q27. Answer: A — Decrease CMR and CMO₂ but increase CBF (directly acting cerebral vasodilator) and hence ICT

Opioids decrease CMR and CMO₂ but increases CBF (directly acting cerebral vasodilator) and hence ICT. This increase in CBF is significant when blood brain barrier is disrupted.

Topic: Pharmacology

Q28. Answer: D — Up to 60%

Ketamine increases CBF and ICT very significantly (up to 60%). Ketamine also increases CMR and CMO₂.

Topic: Pharmacology

Q29. Answer: A — Increases ICT

Succinylcholine increases the ICT.

Topic: Pharmacology

Q30. Answer: B — No, they do not cross the blood brain barrier

Non-depolarizing muscle relaxants do not cross the blood brain barrier.

Topic: Pharmacology

Q31. Answer: B — Decreases CBF, CMR, CMO₂ (but at toxic doses can induce seizure and increase CMR)

Lignocaine decreases CBF, CMR, CMO₂ (but at toxic doses it can induce seizure and can increase CMR).

Topic: Pharmacology

Q32. Answer: C — Increase CBF indirectly by increasing cerebral perfusion pressure

Vasopressors increase CBF indirectly by increasing cerebral perfusion pressure.

Topic: Pharmacology

Q33. Answer: A — Hyperglycemia can precipitate cerebral edema

As hyperglycemia can precipitate cerebral edema therefore, Glucose containing solutions should be avoided.

Topic: General Considerations

Q34. Answer: B — Alternate ringer and normal saline liter by liter

The best policy is to alternate ringer and normal saline liter by liter. Ringer lactate is hypoosmolar and large quantities can aggravate cerebral edema. Normal saline in large quantities can cause hyperchloremic metabolic acidosis.

Topic: General Considerations

Q35. Answer: B — No, routine use of hypothermia is not recommended

Routine use of hypothermia is not recommended.

Topic: General Considerations

Q36. Answer: C — 140-180 mg%

Both hyperglycemia as well as hypoglycemia can be detrimental therefore, blood sugar should be best maintained between 140 and 180 mg%.

Topic: General Considerations

Q37. Answer: A — Capnography

Most important is Capnography. Hypercapnia is the most potent cerebral vasodilator therefore, can significantly increase the ICT.

Topic: General Considerations

Q38. Answer: D — 25-30 mm Hg (moderate hypocapnia)

Moderate hypocapnia (pCO₂ between 25 and 30 mm Hg) is beneficial by decreasing ICT however excessive hypocapnia (pCO₂ < 25 mm Hg) can cause cerebral ischemia.

Topic: General Considerations

Q39. Answer: D — It increases ICT by causing cerebral vasodilation

Hypoxia should be avoided as it increases ICT by causing cerebral vasodilation.

Topic: General Considerations

Q40. Answer: A — Emergence free of coughing, straining and hypertension

Smooth emergence, i.e. emergence free of coughing, straining, and hypertension is must in neurosurgical patients as these can increase ICT and cause massive bleeding.

Topic: General Considerations

Q41. Answer: C — Inject lignocaine before extubation, continue inhalational (at least N2O) till dressing and use beta blockers

The best possible methods to achieve smooth emergence is to inject lignocaine before extubation, continue inhalational (at least N2O) till dressing is over and use beta blockers (Esmolol, Labetalol) if necessary.

Topic: General Considerations

Q42. Answer: C — 70-100 mm Hg

The aim should be to maintain CPP (MAP-ICP) between 70 and 100 mm Hg not only in intraoperative period but also in post-operative period for at least 2-3 days.

Topic: Raised ICP

Q43. Answer: A — They are absolutely contraindicated

Central neuraxial blocks are absolutely contraindicated in raised ICP.

Topic: Raised ICP

Q44. Answer: B — Propofol

Although Thiopentone has more favorable effect on CMR and ICP than propofol but still due to shorter half-life propofol is preferred over thiopentone for induction.

Topic: Raised ICP

Q45. Answer: A — Intubation can significantly increase ICP due to response to laryngoscopy

Intubation can significantly increase the ICP therefore, response to laryngoscopy and intubation must be blunted with lignocaine, adequate depth and beta blockers.

Topic: Raised ICP

Q46. Answer: C — Not absolutely contraindicated - can be safely used for rapid sequence induction or difficult airway

Suxamethonium is not absolutely contraindicated for neurosurgery and can be safely used if there is clear indication like rapid sequence induction or possibility of difficult airway.

Topic: Raised ICP

Q47. Answer: D — IV agents only - inhalational agents should be avoided

If ICP is too high to be controlled or surgical field is too tight that inhalational agents should be avoided and anesthesia is maintained on IV agents.

Topic: Raised ICP

Q48. Answer: D — All patients should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise

All patients of head injury should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise.

Topic: Head Injury

Q49. Answer: D — Unless basal skull fracture has been ruled out

Nasal intubation should not be performed unless basal skull fracture has been ruled out.

Topic: Head Injury

Q50. Answer: A — Modified Glasgow Coma Scale

The best method for assessing neurologic status is modified Glasgow Coma scale.

Topic: Head Injury

Q51. Answer: B — 15/15

Normal score is 15/15. Severe head injury is stated to be present if score is 7 or less.

Topic: Head Injury

Q52. Answer: C — More than 35%

Severe head injury is stated to be present if score is 7 or less and if persist for more than 6 hours is associated with more than 35% mortality.

Topic: Head Injury

Q53. Answer: D — Due to release of brain thromboplastin

DIC is commonly associated with head injury due to release of brain thromboplastin and it must be ruled out in preoperative period.

Topic: Head Injury

Q54. Answer: D — Transesophageal echocardiography (TEE)

The most sensitive tool to detect VAE is transesophageal echocardiography (TEE). It can detect as low as 0.25 mL of air.

Topic: Posterior Fossa Surgery

Q55. Answer: C — > 100 mL

VAE can become clinically significant if the volume of entertained air > 100 mL (however death has been reported with as low as 5 mL of air).

Topic: Posterior Fossa Surgery

Q56. Answer: A — 20-40%

The incidence of venous air embolism (VAE) can be as high as 20-40% in posterior fossa craniotomies however, fortunately the incidence of clinically significant VAE is very less.

Topic: Posterior Fossa Surgery

Q57. Answer: B — Stop nitrous oxide and start 100% oxygen (N2O can expand the size of air embolus)

Stop nitrous oxide and start 100% oxygen. Nitrous oxide can expand the size of air embolus.

Topic: Posterior Fossa Surgery

Q58. Answer: D — TEE > Precordial Doppler > PAP, ETCO2, ET nitrogen > CVP > ECG > Clinical signs

The sensitivity in decreasing order is: TEE > Precordial Doppler > PAP, ETCO2, ET nitrogen > CVP > ECG > clinical signs.

Topic: Posterior Fossa Surgery

Q59. Answer: A — Hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse

Clinical signs include hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse. Clinical signs appear late in venous air embolism.

Topic: Posterior Fossa Surgery

Q60. Answer: A — 4-14 days

After stroke surgery should be avoided in period of maximal vasospasm risk (4-14 days) as far as possible.

Topic: Stroke/Cerebral Aneurysms